SUHAS NAGARAJ

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EDUCATION

Master of Engineering in Robotics [Minor: Artificial Intelligence] [CGPA: 4.0 / 4.0] **August 2023 - May 2025**
University Of Maryland, College Park [Phi Kappa Phi Honor Society Member] College Park, United States

Bachelor of Engineering in Mechanical Engineering [CGPA: 9.3 / 10.0] **August 2017 - July 2021**
Ramaiah Institute of Technology Bengaluru, India

PROFESSIONAL EXPERIENCE

Teaching Assistant – MSML605 Computing Systems for Machine Learning **January 2024 – Present**
College of Computer, Mathematical & Natural Sciences, University of Maryland College Park, USA

- Assisting Dr. Jerry Wu in teaching a comprehensive course on Python programming fundamentals and hardware aspects of machine learning systems, including computer architecture, parallel processing, and high-performance computing technologies crucial for implementing and optimizing machine learning algorithms.

Research Intern **May 2024 – Present**
Intelligent Inclusive Interaction Design (I3D) Lab, Indian Institute of Science (IISc) Bengaluru, India

- Conducted Research & Development work under Dr. Pradipta Biswas (Robert Bosch Centre for Cyber Physical Systems)
- Developed a novel 3D reconstruction algorithm integrating LiDAR sensors with 3D Gaussian Splatting SLAM and ROS2, achieving a 200% improvement in localization accuracy (measured as RMSE) and a 35% reduction in computational load, significantly enhancing autonomous navigation and mapping capabilities; Worked with Turtlebot3 Waffle, Tortoisebot and Tankbot mobile robots for integration

Engineering Intern **October 2022 – March 2023**
DiFACTO Robotics and Automation Bengaluru, India

- Excelled in designing automation parts using SolidWorks; conducted industrial process simulations with FANUC RoboGuide; Gained proficiency in offline and online programming of industrial robots (ABB RAPID and FANUC KAREL); Contributed to the assembly of Automated Guided Vehicles deployed in the Maruti Suzuki assembly line.
- Developed skills in PLC programming and Devicenet interfacing to effectively control robot cells using Mitsubishi PLC.

Technical Consultant **August 2021 – March 2022**
U-Solar Clean Energy Solutions Pvt. Ltd. Bengaluru, India

- Managed more than 50 client and 10 partner accounts, accelerated sales as part of the inaugural team of Technical Consultants, contributing to a 20% increase in customer retention; Spearheaded market and policy research initiatives, resulting in a 15% growth in market reach. Devised and implemented strategies that drove increase in leads and projects in the pipeline

PUBLICATION

3D Reconstruction via Camera-Lidar (2D) Fusion for Mobile Robots: A Gaussian Splatting Approach
*Ajay Kumar Sandula, *Shriram Damodaran, ***Suhas Nagaraj***, Debasish Ghose and Pradipta Biswas [* are equally contributing authors]*
IEEE International Conference on Robotics and Automation (ICRA) 2025

SKILLS

- **Technical Expertise:** ROS1/ROS2, Computer Vision, Machine Learning, Neural Networks, Natural Language Processing, Simulation, Robot Programming, SLAM, 3D reconstruction, Data Analytics, Sensor Fusion, Multi-threading, CUDA, Robot Hardware and Modelling, Could Computing, MLOps, Containerization, Version Control, Optimization, Control theory, Product Design, Rapid Prototyping, PLC Programming.
- **Programming Languages:** Python, C++, MATLAB, RAPID (ABB), KAREL (FANUC).
- **Libraries, Frameworks and Tools:** PyTorch, Cuda, Tensorflow, Keras, ONNX, Huggingface, OpenCV, Open3D, PCL, Yolo, Cartographer, NumPy, pandas, scikit-learn, NLTK, Kubernetes, Docker, AWS, Azure, GCP, Linux, GitHub.
- **Others:** MATLAB Simulink, FANUC ROBOGUIDE, ABB Robot Studio, CimStation Robotics, SolidWorks, AutoCAD, Fusion 360, CATIA.

PROJECTS [www.suhasnagaraj.com/projects]

- **Adaptive Text-to-Command Translation for Robotic Navigation:** Fine-tuned T5-Small and LoRA models to translate natural language commands into structured navigation plans for TurtleBot3 in ROS2 and Gazebo
- **ASD Prediction Using Eye Tracking Data:** Developed a novel deep learning approach (CNN, LSTM, Transformers) for classifying Autism Spectrum Disorder using eye-tracking data, achieving 88.9% accuracy.
- **Automated Guided Vehicle:** Designed autonomous navigation with lane following, stop sign detection, and dynamic obstacle avoidance for TurtleBot3 using ROS2 and OpenCV in simulation and real-world setups.
- **Maze Navigation with Object Detection:** Developed ROS2-based system for maze traversal using Aruco markers and object detection
- **SLAM-Based Waypoint Navigation:** Designed an autonomous waypoint navigation system for TurtleBot3 using SLAM, TF tree localization.
- **Deep Reinforcement Learning:** Applied Deep Q-Networks (DQN) for obstacle avoidance and autonomous navigation in a Turtlebot3.
- **Path Planning Algorithms:** Developed Dijkstra's, A*, RRT, RRT*, Bidirectional A*, Potential Field and Real Time RRT* algorithms for mobile robot navigation, implemented the solutions on a Turtlebot3 Waffle robot.